Diffusion Model for Image Inpainting

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***Abstract*—This project will develop a state-of-the-art, mask-conditioned diffusion inpainting system capable of reconstructing missing or corrupted image regions with high semantic fidelity and seamless boundaries. Building on latent diffusion and classifier-free guidance, we will (i) engineer robust mask and context conditioning, (iiii) introduce boundary-aware reconstruction losses, and (iii) quantify performance across mask shapes/sizes and semantic control settings (with/without text prompts). The result will be an open, reproducible pipeline with strong baselines, rigorous ablations, and a deployable demo.**

***Index Terms*—Image inpainting; Diffusion models; Latent diffusion; Mask-conditioned generation; Text-guided inpainting; Classifier-free guidance; Few-step sampling (DDIM/DPM-Solver); LoRA fine-tuning; Boundary-aware loss; Perceptual loss (LPIPS); PSNR; SSIM; FID; KID; Poisson/alpha blending; Semantic coherence; Robustness to mask size/shape; Reproducibility.**

I. Introduction

Image inpainting aims to reconstruct missing or corrupted regions so that the completed image looks natural and consistent with the visible context. Classical patch-based and PDE methods can copy texture and continue edges, but they fail on large holes or complex semantics. CNN/GAN models improved plausibility, yet often suffer from blurry textures, repetitive patterns, and visible seams where the fill meets the original pixels.

Diffusion models—particularly latent diffusion—now set the standard for controllable, high-fidelity image synthesis. By learning a powerful generative prior and allowing flexible conditioning (masks, text, edges, depth), they can generate content that matches global scene semantics while preserving local texture detail. However, practical, reliable inpainting still faces challenges: edge blending, prompt drift, large-hole realism, runtime, and reproducibility. This project targets those pain points with a mask-aware diffusion pipeline that combines strong conditioning, boundary-aware objectives, and rigorous evaluation.

A. Background & Motivation

Image inpainting is used to remove objects, restore damaged photos, and edit images without visible seams. Classic methods (e.g., patch-based or diffusion-of-pixels in the PDE sense) often fail on large holes or complex textures. Deep CNN models improved the quality but may struggle with global coherence. Modern diffusion models (DDPM/latent diffusion) generate high-fidelity content and can be conditioned on masks and prompts, making them strong candidates for realistic, controllable inpainting. This project leverages those strengths for a practical, reproducible pipeline.

B. Problem Statement

Given an image and a binary mask (1 = hole to fill, 0 = keep), synthesize such that:

- The filled region is realistic and semantically consistent with the visible context.
- Boundaries between filled and original pixels are seamless.
- (Optional) The content follows a text prompt if provided.

C. Scope and Assumptions

Scope:

- **In-scope:** Single-image inpainting with a binary mask on natural images at $\sim$ 512 $\times$ 512; latent-diffusion with mask conditioning (optional text prompts); LoRA fine-tuning; light alpha/Poisson post-blend; baselines (PatchMatch, PDE, LaMa/EdgeConnect); metrics (PSNR, SSIM, LPIPS, FID/KID, boundary score, CLIPScore).
- **Out-of-scope:** Video/multi-view/3D, identity-preserving face edits, training from scratch, non-image tasks, deceptive/forensic/NSFW use, mobile/on-device or hard real-time guarantees.

Assumptions:

- Licensed datasets and base checkpoints; LoRA adapters may be released.
- $\geq$ 1 GPU with $\sim$ 16–24 GB VRAM; PyTorch + `diffusers` stack available.
- Metrics above are acceptable proxies; a small (non-IRB) user study if used.
- Prompts are optional and safe; guidance is applied only inside the mask.
- Reproducible runs (fixed seeds/configs); demo shows disclosure/watermark.
- Primary target is 512 $\times$ 512; higher resolutions are best-effort.

II. Literature Review

A. Existing Work

- **Classical methods:** PatchMatch, PDE-based (Telea /Navier–Stokes). Fast and simple, good for small

holes and smooth backgrounds, weak for large missing areas and complex objects.

- **CNN/GAN inpainting:** Gated/partial convolutions (e.g., DeepFillv2), structure-guided models (e.g., EdgeConnect), and large-mask CNNs (e.g., LaMa). Better context reasoning than classical, but still visible seams, texture repetition, and limited semantic control.
- **Transformer/priors:** Vision transformers and priors (edges, segmentation, depth) help with structure, yet training stability and global coherence remain issues on large masks.
- **Diffusion inpainting:** DDPM/latent-diffusion with mask conditioning (e.g., RePaint-style iterative denoising; SD-inpainting variants). These achieve state-of-the-art perceptual quality and accept text guidance but can struggle with boundary smoothness, prompt faithfulness, speed (many steps), and high-resolution deployment.

B. Research Gaps (Related Work)

Seams at boundaries; prompt drift; large-hole semantics; weak structure preservation; efficiency vs. quality; mask-shape sensitivity; underused edge losses; metric blind spots; weak reproducibility/governance; fragile hi-res deployment.

C. Research Aim

We aim to build a mask-conditioned latent-diffusion inpainting system with seamless boundaries, strong large-mask semantics, reliable (optional) prompt control, and efficient inference at practical resolutions.

D. Intended Contributions

- Boundary-aware training with narrow-band loss + gentle compositing.
- Robust conditioning (mask + masked image; optional structure; masked-region CFG).
- Practical efficiency (few-step sampling, LoRA fine-tuning, lower VRAM/latency).
- Rigorous evaluation (boundary metrics, CLIPScore, robustness sweeps, blinded user study).
- Reproducibility & governance (public configs/seeds, mask generators, model cards, safe-use notes).

III. Objectives and Key Results

Objectives:

- **O1:** Design a diffusion-based inpainting model with explicit mask/context conditioning.
- **O2:** Achieve superior perceptual quality and boundary coherence vs. strong baselines.
- **O3:** Provide text-guided control (optional prompt) without degrading fidelity.
- **O4:** Deliver a reproducible codebase, pretrained weights (subject to license), and a demo.

Key Results (KRs):

- **KR1:** $\geq 10\%$ LPIPS$\downarrow$ improvement and $\geq 3\%$ SSIM$\uparrow$ vs. best deep baseline on ≥ 2 datasets.
- **KR2:** Smoothness at mask edges: $\geq 15\%$ reduction in boundary gradient error.
- **KR3:** Robustness across mask area buckets ($\leq 10\%$, 10–30%, $\geq 30\%$) with $\leq 5\%$ relative drop in LPIPS.
- **KR4:** Ablation table isolating contributions from guidance, schedulers, and boundary loss.

IV. Proposed Methodology

A. Data

- **Use:** Widely adopted datasets for inpainting (e.g., Places2 / COCO-2017 subsets or similar).
- **Create:** Training masks procedurally (rectangles, strokes, irregular shapes) and include realistic masks (object removal).
- **Split:** train/val/test with no scene leakage.

B. Preprocessing

- Resize/crop to a fixed resolution (e.g., 512×512).
- Normalize images; generate binary masks; create masked inputs.

C. Model

- **Backbone:** Pre-trained latent diffusion (Stable-Diffusion–style UNet in latent space).
- **Conditioning:** Concatenate the mask and masked image as inputs. Optionally include text embeddings (classifier-free guidance).
- **Objective:** Predict noise with standard diffusion loss; add reconstruction loss on the filled region (L_1 + perceptual loss) to stabilize.
- **Sampling:** DDIM or PNDM scheduler; guidance scale sweep to balance realism vs faithfulness.
- **Variants (Ablations):**
  - w/ vs w/o text prompts,
  - different schedulers/steps,
  - boundary refinement loss (blend around mask edges),
  - lightweight LoRA fine-tuning vs full fine-tuning.

D. Baselines

- **Traditional:** OpenCV Telea / Navier-Stokes, PatchMatch.
- **Deep models:** A strong prior CNN inpainting model (e.g., LaMa/EdgeConnect or similar) for context.

E. Training & Implementation

- **Framework:** PyTorch + `diffusers` (Hugging Face).
- **Strategy:** Start from a pre-trained latent diffusion checkpoint; fine-tune on masked images; use mixed precision; early stopping on val LPIPS.
- **Compute:** $1\times$ GPU with $\geq$ 16–24 GB VRAM recommended; gradient accumulation if needed.

F. Evaluation

- **Quantitative:**
  - PSNR/SSIM on masked region,
  - LPIPS (perceptual similarity),
  - FID on inpainted crops,
  - Masked-L_1 (error only where $M = 1$).
- **Robustness:** Performance vs mask area (e.g., 10%, 20%, 40% of image), irregular vs rectangular masks, text-prompt consistency (CLIPScore if prompts used).
- **Qualitative:** Side-by-side examples; small user study (n $\approx$ 15–20) rating realism (Likert 1–5).
- **Runtime:** Average seconds per image at set resolution and steps.

V. Experimental Plan

- **Baseline Setup:** Run PatchMatch/Telea and a CNN inpainting model; log metrics.
- **Diffusion Zero-Shot:** Mask-conditioned inference without fine-tuning.
- **Fine-Tuned Model:** Train with reconstruction + perceptual loss; compare.
- **Ablations:**
  - mask shape/size sweep,
  - scheduler and step counts,
  - guidance scale sweep,
  - LoRA vs full fine-tune.
- **Text Guidance:** Prompt vs no-prompt on scenes where semantic control matters.
- **User Study:** Blind rating of realism and boundary quality.

VI. Risks & Mitigations

- **Artifacting at edges:** Boundary loss + post-blend; tune mask dilation margin; latent-space patch consistency.
- **Semantic drift with prompts:** CFG scale search; negative prompts; partial CFG (masked-region-only).
- **Slow inference:** DPM-Solver/F; distillation to few-step; attention kernels; ONNX/TensorRT export (if feasible).
- **Data bias / misuse:** Dataset audits, failure-case log; content filters; watermarking of outputs; explicit terms of use.

VII. Ethical Considerations

- Prevent harmful edits (e.g., deceptive photo manipulation).
- Address dataset bias; report failure cases across demographics and scenes.
- Provide a watermark toggle and a clear disclosure in the demo UI.

VIII. Deliverables

- Code repository with training, evaluation, and inference scripts.
- Pretrained/Fine-tuned weights (if license permits).
- Demo app (Gradio) for mask upload and optional text prompts.
- Report (6–10 pages) with methods, experiments, results, and limitations.
- Dataset/mask generation scripts and evaluation notebook.

IX. Timeline (10–12 Weeks)

- **W1:** Finalize dataset splits, mask generator, baselines.
- **W2–W3:** Integrate diffusion pipeline, zero-shot inference; metric harness.
- **W4–W6:** Fine-tune (LoRA first), validate, pick best configs.
- **W7–W8:** Ablations (masks, schedulers, steps, guidance).
- **W9:** Text-guided experiments; CLIPScore analysis.
- **W10:** User study, qualitative gallery, runtime profiling.
- **W11–W12:** Write report, polish demo, prepare submission.

X. Resources & Budget (indicative)

- **Compute:** 1–2× 24–48 GB GPUs for training; 12–16 GB GPU sufficient for inference; $\sim$ 200–400 GPU-hours for adaptation + sweeps.
- **Software:** PyTorch, Hugging Face `diffusers`/`transformers`, OpenCV, `torchmetrics`, Gradio.
- **Personnel:**
  - 1× Lead Research Engineer (architecture/algorithms),
  - 1× Applied Engineer (data/eval/demo),
  - 0.5× MLE (infra, CI, packaging).

XI. Expected Impact & Limitations

- **Impact:** High-quality, controllable inpainting for restoration, object removal, and editing pipelines.
- **Limitations:** Extreme holes ($>$ 50% area), fine-grained identity preservation without specialized priors, prompt ambiguity; potential failure on rare categories not represented in data.

XII. Conclusion

This proposal advances a practical, high-fidelity solution for image inpainting by coupling mask-conditioned latent diffusion with boundary-aware objectives and efficient few-step sampling. The approach targets the core pain points observed in prior work—seam artifacts, semantic drift on large or irregular masks, fragile prompt control, and high

runtime—while maintaining strict reproducibility and governance (licensed data, public configs, seeds, and a model card).

We expect to exceed strong classical and CNN baselines on perceptual and boundary metrics, deliver reliable prompt-guided edits when used, and provide an openly reproducible pipeline with a lightweight demo suitable for desktop GPUs. The work is deliberately scoped to single-image inpainting at practical resolutions, with a clear path to extensions such as structure-aware conditioning, higher-resolution deployment, few-step distillation, and eventually video or multi-view consistency. By combining rigorous evaluation with transparent reporting and safe-use guidance, the project aims to produce a state-of-the-art yet responsible inpainting system that is both research-grade and ready for real-world integration.

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